

「研究者」や「博士」と聞いて何を想像しますか？

- テストこちら



Comfort Intelligence for Human-Robot Interaction

“快適化な知能を有する
ヒューマンロボットインタラクションの実現”



奈良先端科学技術大学院大学
インタラクティブメディア設計学研究室
准教授 澤邊太志
(Prof. Dr. Max Taishi Sawabe)



HRI: Human-Robot Interaction

- "Design interaction between Human to Robot"
- Fields : Engineering, Information Science, Psychology, Sociology, Social Welfare Studies etc.



Comfort means a state of physical or mental ease, free from pain, worry, or suffering.  Facebook · English Learning

You can check more detailed definitions in the [Cambridge Dictionary](#) or explore general concepts on [Wikipedia](#).

Noun Meanings

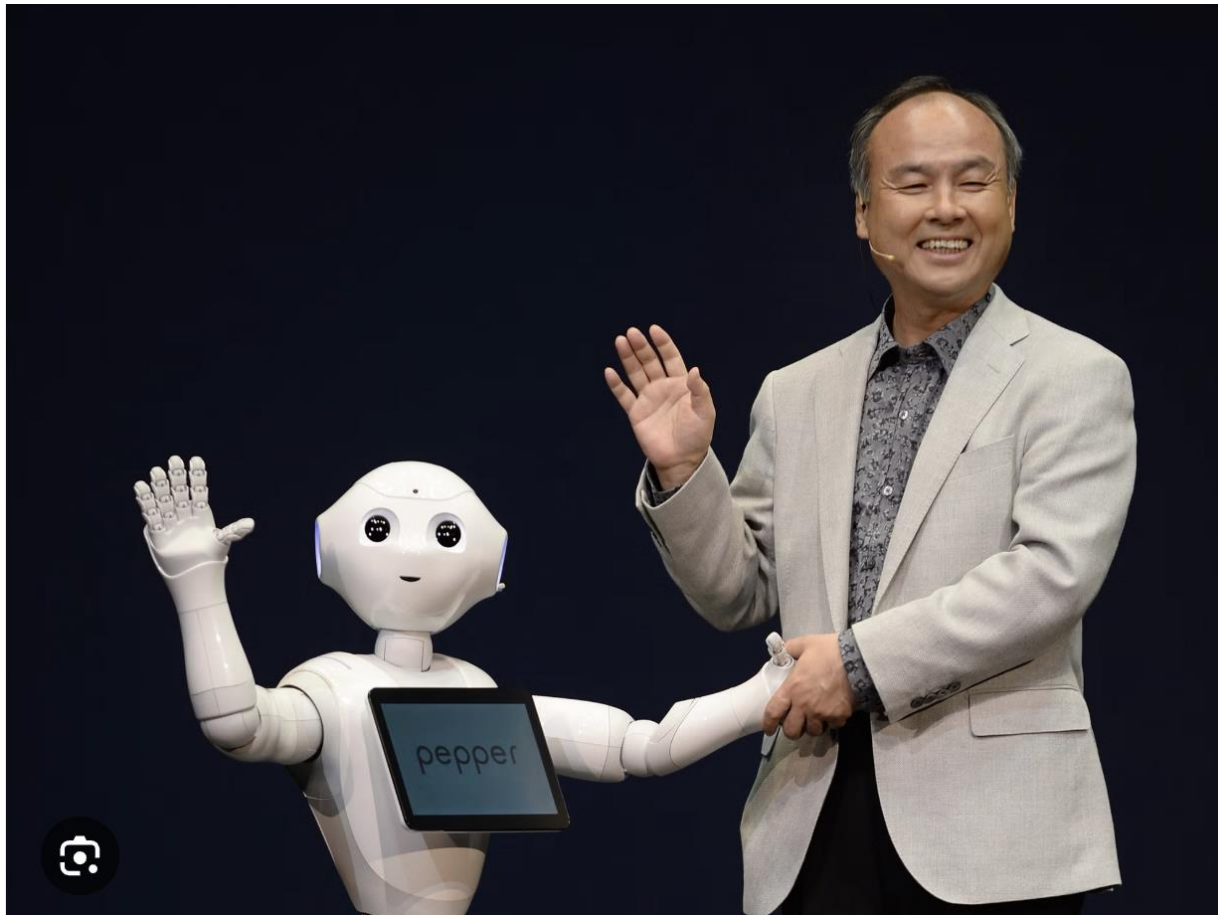
- **Physical ease:** A pleasant feeling of being relaxed with no physical pain.
- **Consolation:** A state of feeling better after sadness or grief, or something that brings relief.
- **Material well-being:** Having enough money and pleasant things to make life easy.  Cambridge Dictionary

Verb Meanings

- **To console:** To make someone who is sad, worried, or upset feel better through kind words or sympathy.  Oxford Learner's Dictiona...

If you'd like, I can give you **synonyms** or **example sentences** for a specific way you want to use the word.

Humanoid robot (before)



Soft Bank Inc. Papper robot with CEO



Papper working at Karaoke

Now....

Pepper's Graveyard



<http://www.all-nationz.com/archives/1079594649.html>

What is Comfort for You (Human)?

HRI (Human-Robot Interaction)

Robots are not just machines.
They are partners in our daily lives.

Small interactions /
create a
bigger future.

Education

Personalized
learning and
new experiences



Support

Companionship
and emotional
support



Healthcare

Supporting
mental and
physical well-being



Work

Collaboration
and productivity



Entertainment

More fun
and immersive
experiences



The HRI Loop

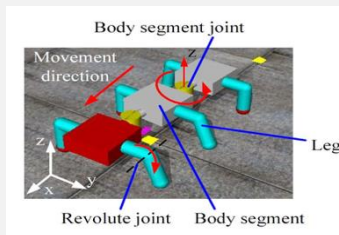


Better
Together

1. NAIST (准教授) :

“Nara Institute of Science and Technology, Associate Professor.”

- Human-Robot Interaction
- AR/VR
- Self Driving Vehicles



Prof. Hirose, Prof. Ma (Biomimetics)



神原雅之准教授

Prof. Hagita, Prof. Kanbara (HRI, Self-driving)



Prof. Kato (AR/VR)



2. amirobo tech Inc. (CEO) :

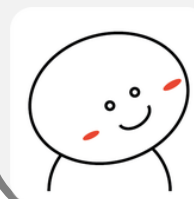
“IT company in Japan.”

(Research based business!!!)

- iOS/Android App dev
- VR/AR App dev
- UI/UX design dev

- Hobby -

Coffee, Museum, Language, **LEGO**,
Basketball, Movie and **ANIME**,
, **Creating and Playing** ...



Enjoy your life!

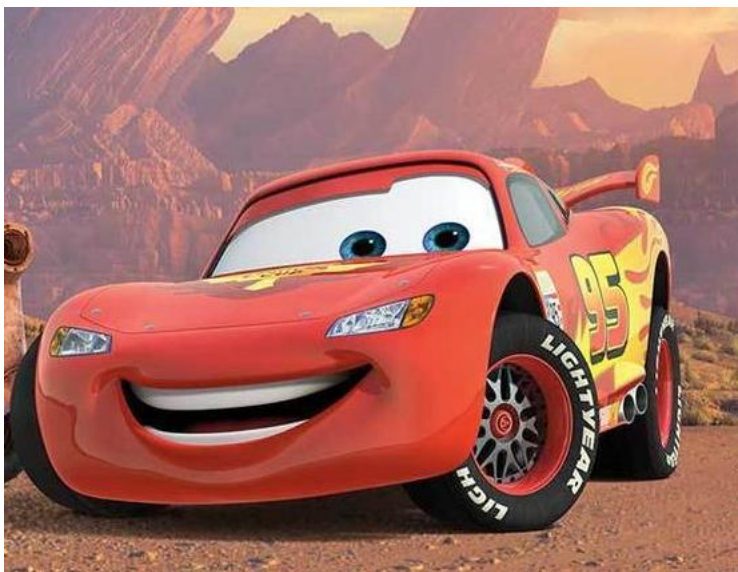
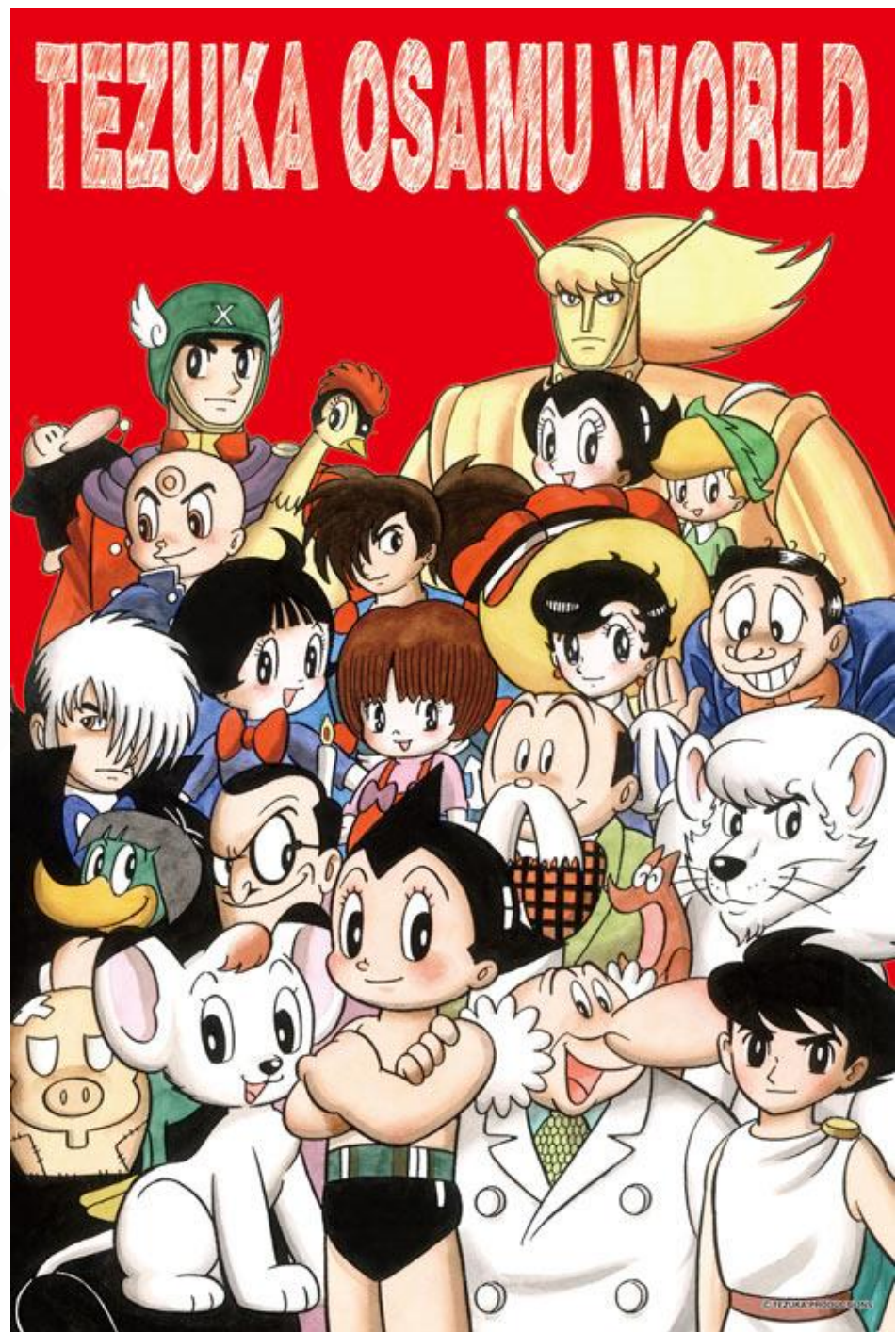
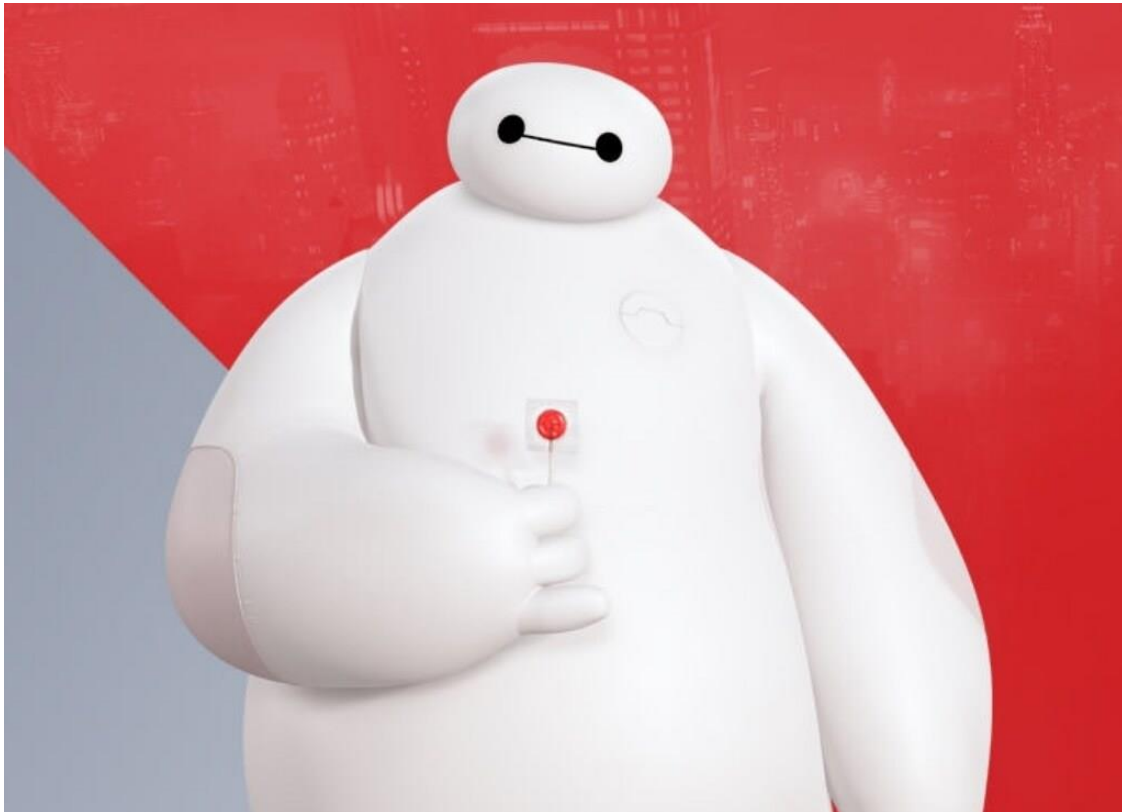
大事なこと : important things

Curiosity & Interaction

好奇心

相互作用

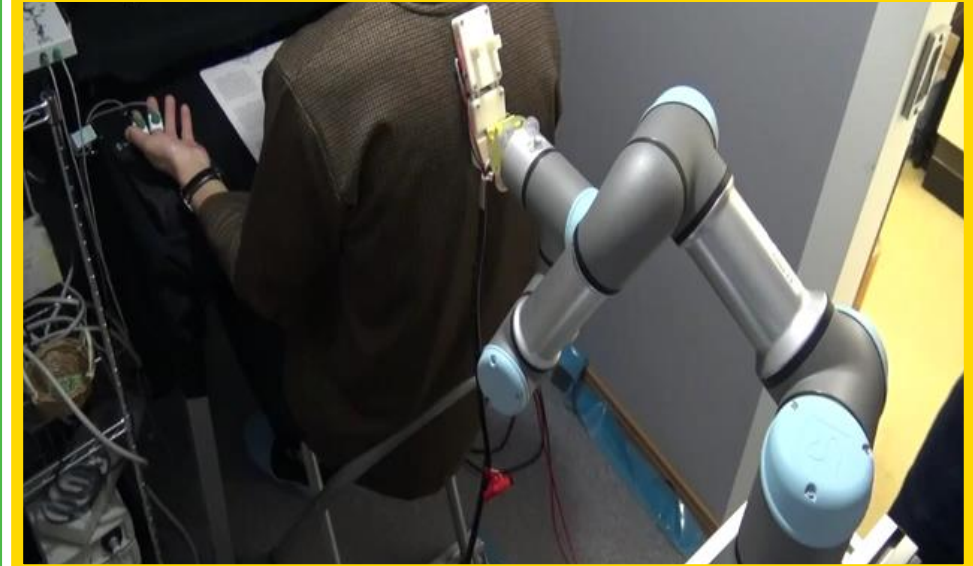
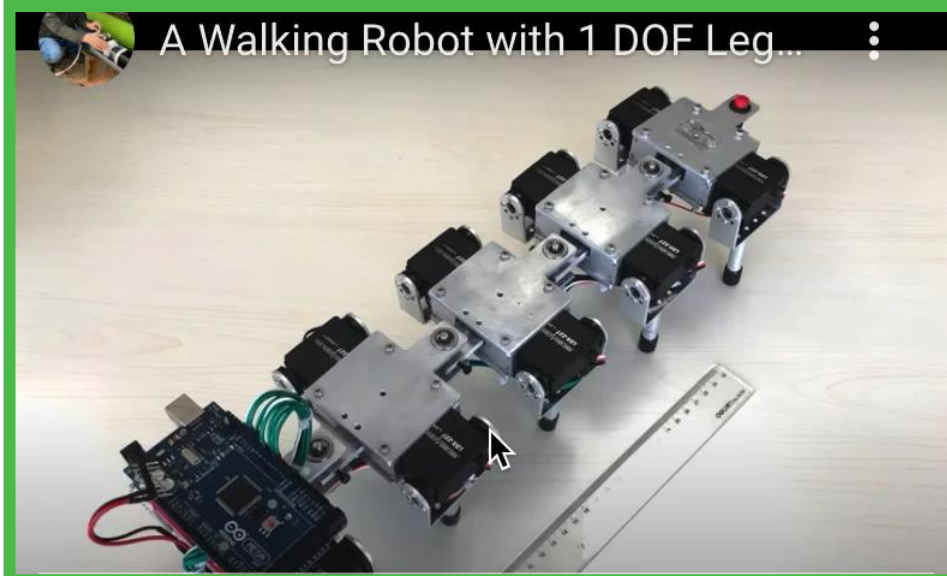




Create Human-Comfortable Interaction

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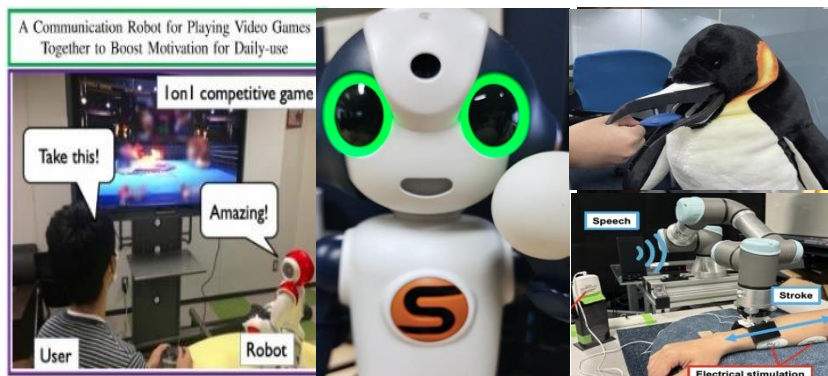
What would be
comfortable -
interaction for
HRI?



Autonomous wheelchairs

Interactive Media Design Lab : インタラクティブメディア設計学研究室 (加藤研)

Our vision is to introduce Augmented Reality & Human-Robot Interaction into the everyday lives of everyone on this planet. Those a technology that enhances human vision with computer generated graphics.



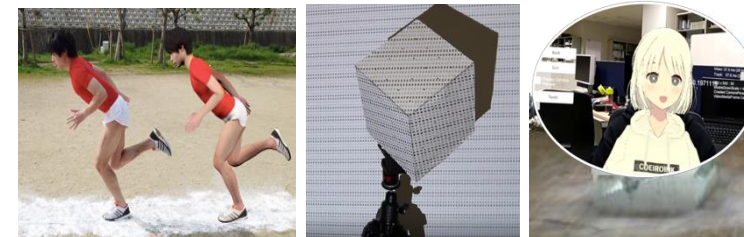
Human-Robot Interaction

対話エージェント、タッチロボット、
快適な自動運転、生命感アニマトロニクス



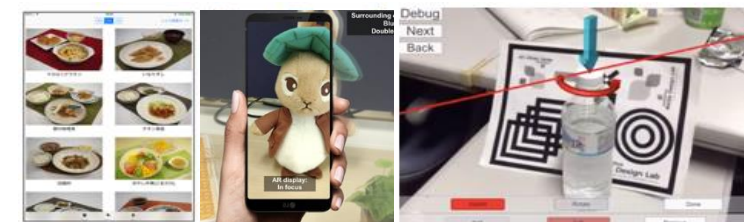
AR/VR (拡張現実感・仮想現実)

ARコンテンツ設計支援、医療AR・VR、
作業支援・学習・訓練



Computer Vision

姿勢推定、スポーツ応用、
デプス推定・統合

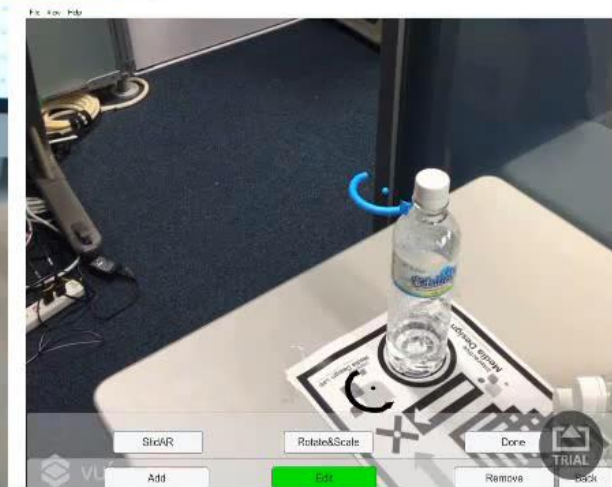
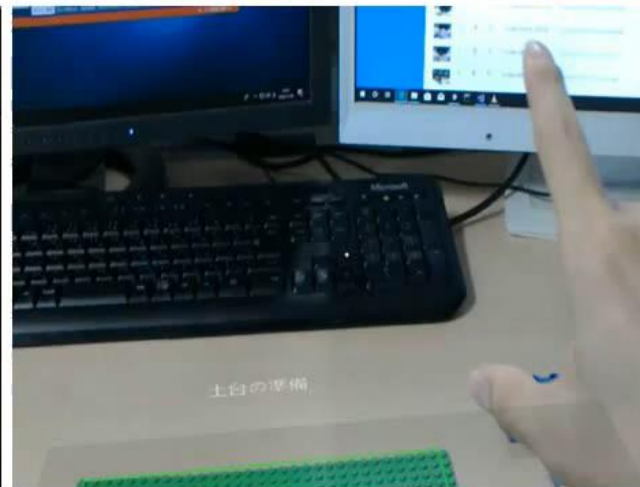
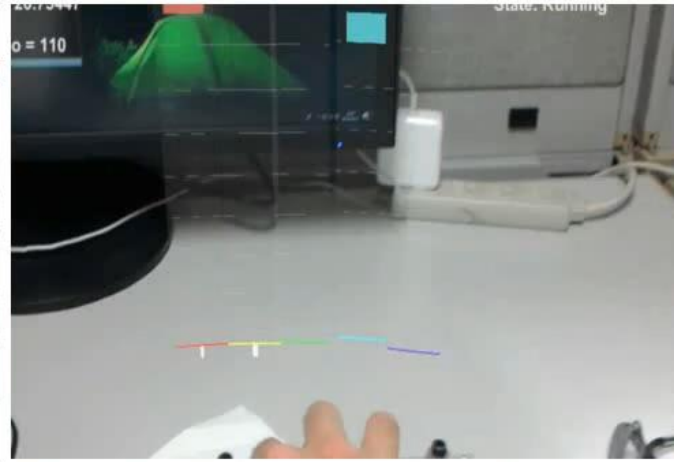
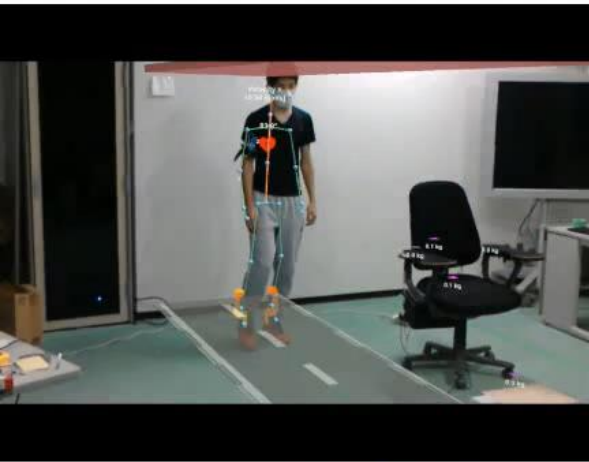


Human-Computer Interaction

認知行動変容、UI/UX
次世代インターフェース開発

・就職先(Career) : SONY、NTT、任天堂 (Nintendo)、IBM、カプコン (Capcom)、Yahoo! Japan、OMRON、Panasonic、楽天、Microsoft、Canon三菱電機、朝日放送テレビ、富士通研究所、日立製作所...

IMD Keywords: AR/VR, HCI, HRI



[Research](#)[Publications](#)[Awards](#)[Infrastructure](#)[Member](#)[Contact](#)[Blog](#)

Human-Computer Interaction

[詳細 ▶](#)

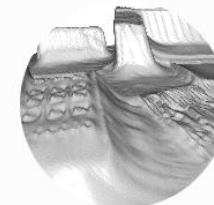
Human-Robot Interaction

[詳細 ▶](#)

Computer Graphics

[詳細 ▶](#)

AR/VR



Computer Vision

[詳細 ▶](#)

設備等

様々なヘッドマウントディスプレイ



Microsoft HoloLens



Apple Vision Pro



MagicLeap



Google glass



HTC Vive Pro



Oculus Quest 2



Varjo XR-1



Sota



DOBOT Magician



Nao



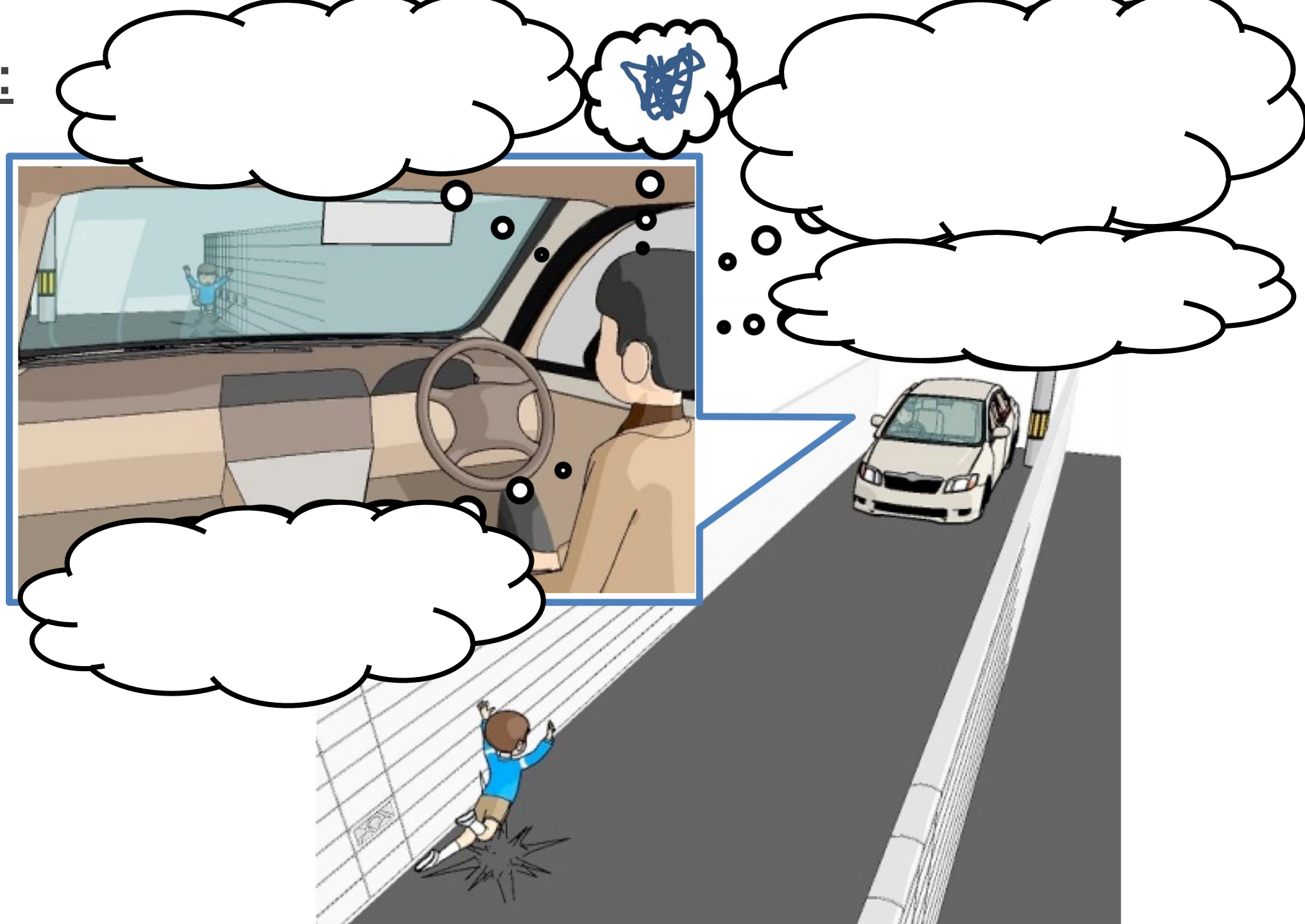
うなずきかぼちゃん



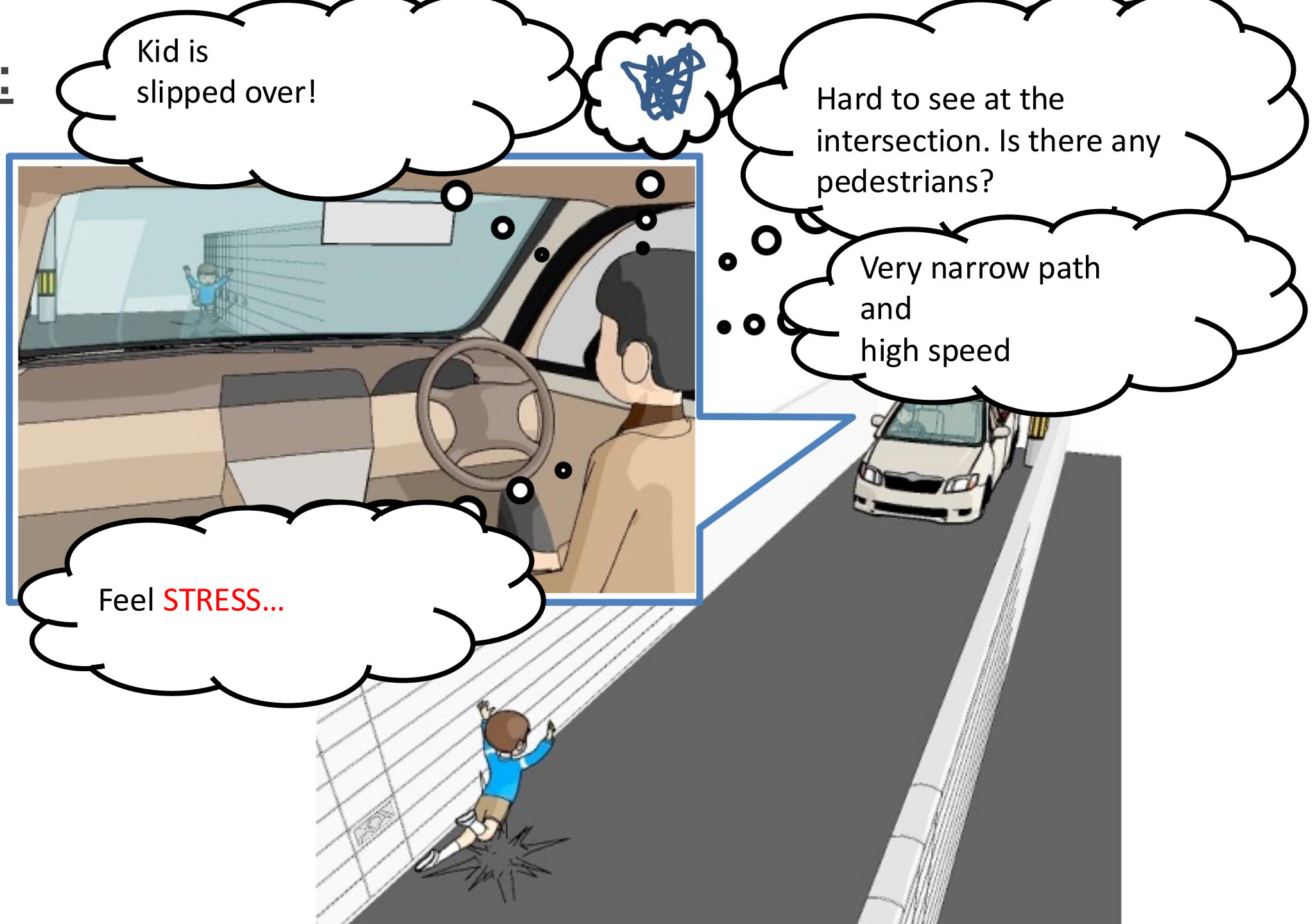
RoboHoN



Example:

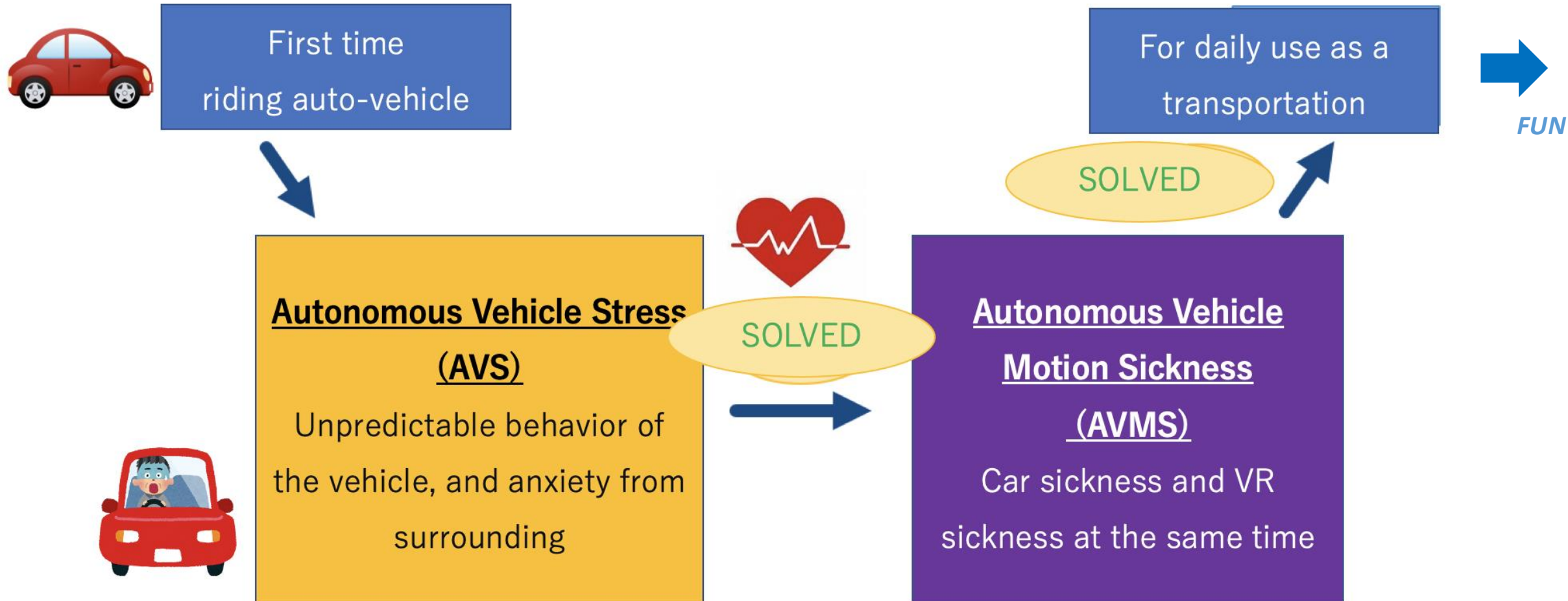


Example:



Comfort Intelligence for Autonomous Vehicles

- Provide enough comfort for all passengers [8]: [8] T. Sawabe et al., 2018





Propose: stress estimation using car camera



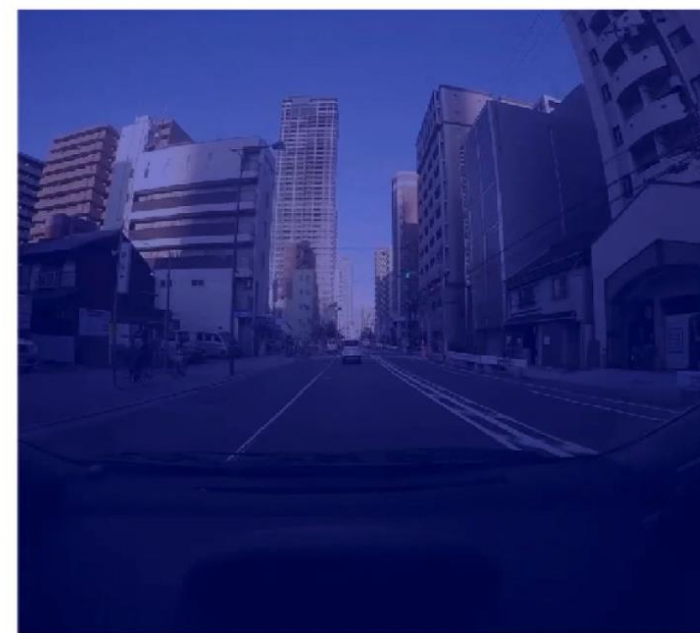
Contents :

Estimate and evaluate the stress level from the car camera

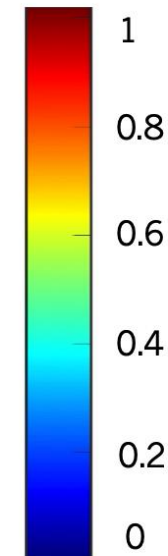
Novelty:

- Different types of stress can be an estimated
- Only using a camera inside the car

[19]. ITS研究会2017



Stress anotation



ストレス確率

Stress level

Result :

Can be estimated stress level

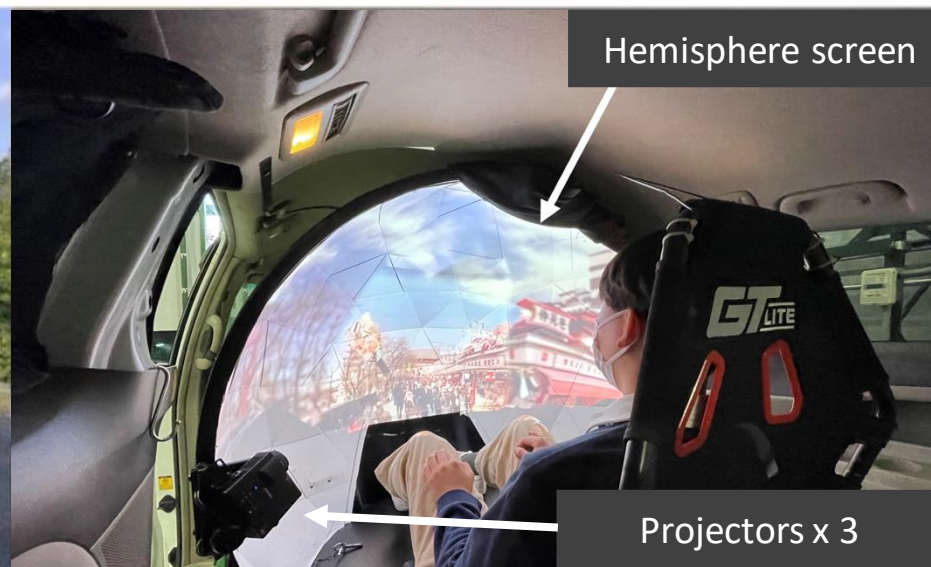
Outline of XR Cabin



Exterior of autonomous vehicle

1. Autonomous vehicle system based on Autoware.AI*

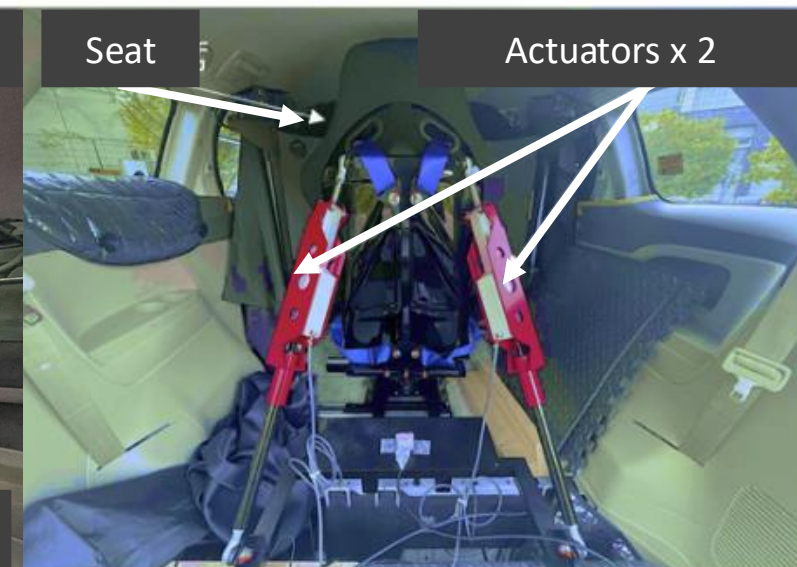
* Acceleration, braking, steering control can be automated



Immersive hemisphere screen

2. Immersive VR/AR information display system*

* Projection based cylinder type screen
* Head Mounted-Display (HMD)



Motion platform seat

3. Motion platform seat system*

* The seat can be controlled and adjusted at a specific angle.
(0 – 150 mm)

XR Cabin: autonomous vehicle functions

- The system can follow the same trajectory at the same speed over and over.

Exterior:



Interior:

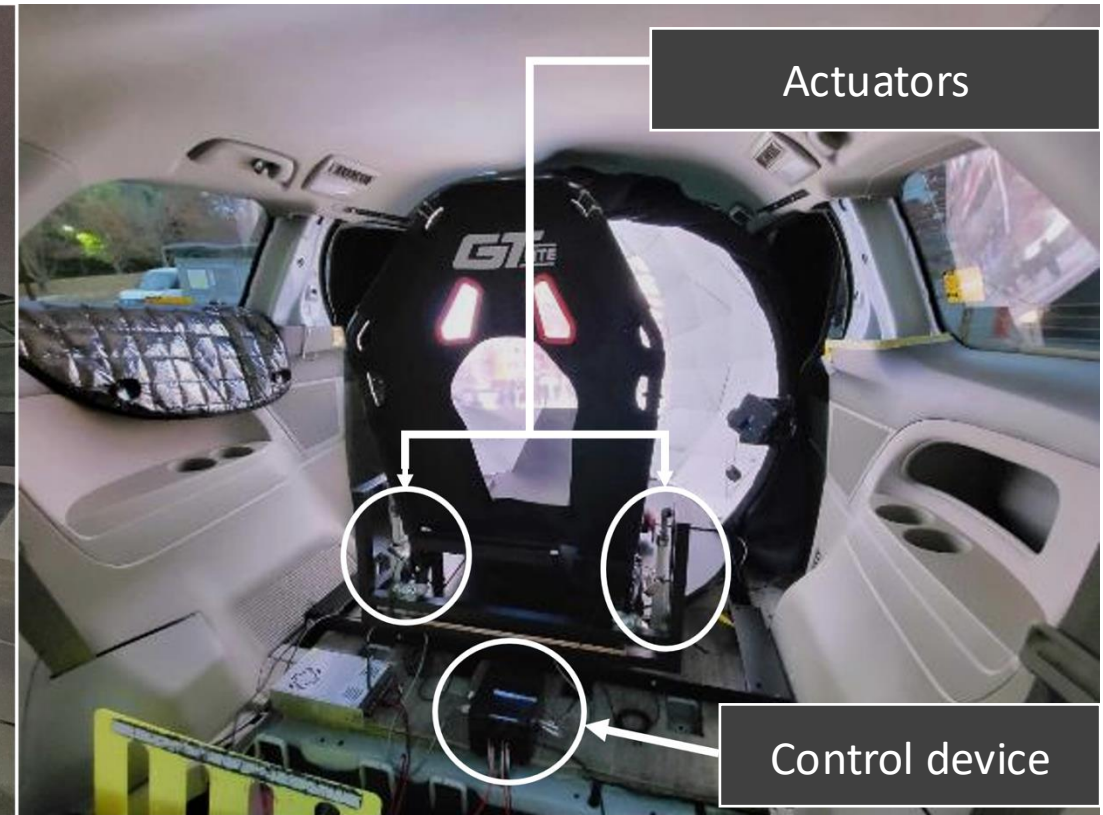


XR Cabin: Outline of immersive screen and motion platform seat

- Systems to control the sense of movement.



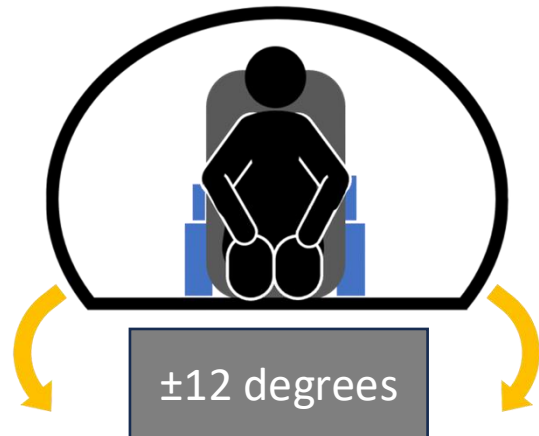
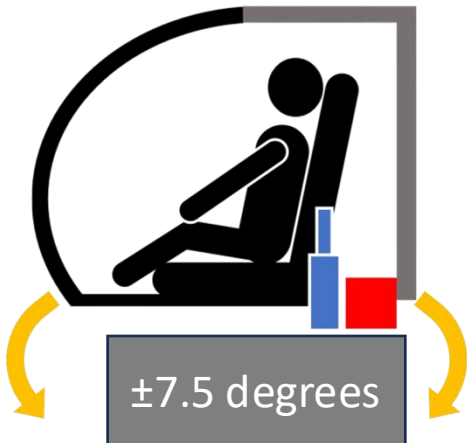
Immersive hemisphere screen



Motion platform seat

XR Cabin: Details of immersive screen and motion platform seat

- The hemisphere screen and motion platform seat **are combined together** and can be tilted.
- **Motion platform seat:**
 - >Maximum expansion and contraction: 150mm,
 - >Expansion and contraction speed: 10mm/s,
 - >Maximum load: 800 N,
 - >Range of motion: 15 degrees forward and backward, 24 degrees right and left side.



○Takuto Akiyoshi, Masashi Abe, Yuki Shimizu, Yusaku Takahama,
Kouki Nagata, Yosuke Okami, Taishi Sawabe
Nara Institute of Science and Technology (NAIST), Japan

- “*Hype Dlive*” is an XR live system for an autonomous vehicle that uses unpredictable stressful behavior to make passengers excited without noticing



[XR live performance content]

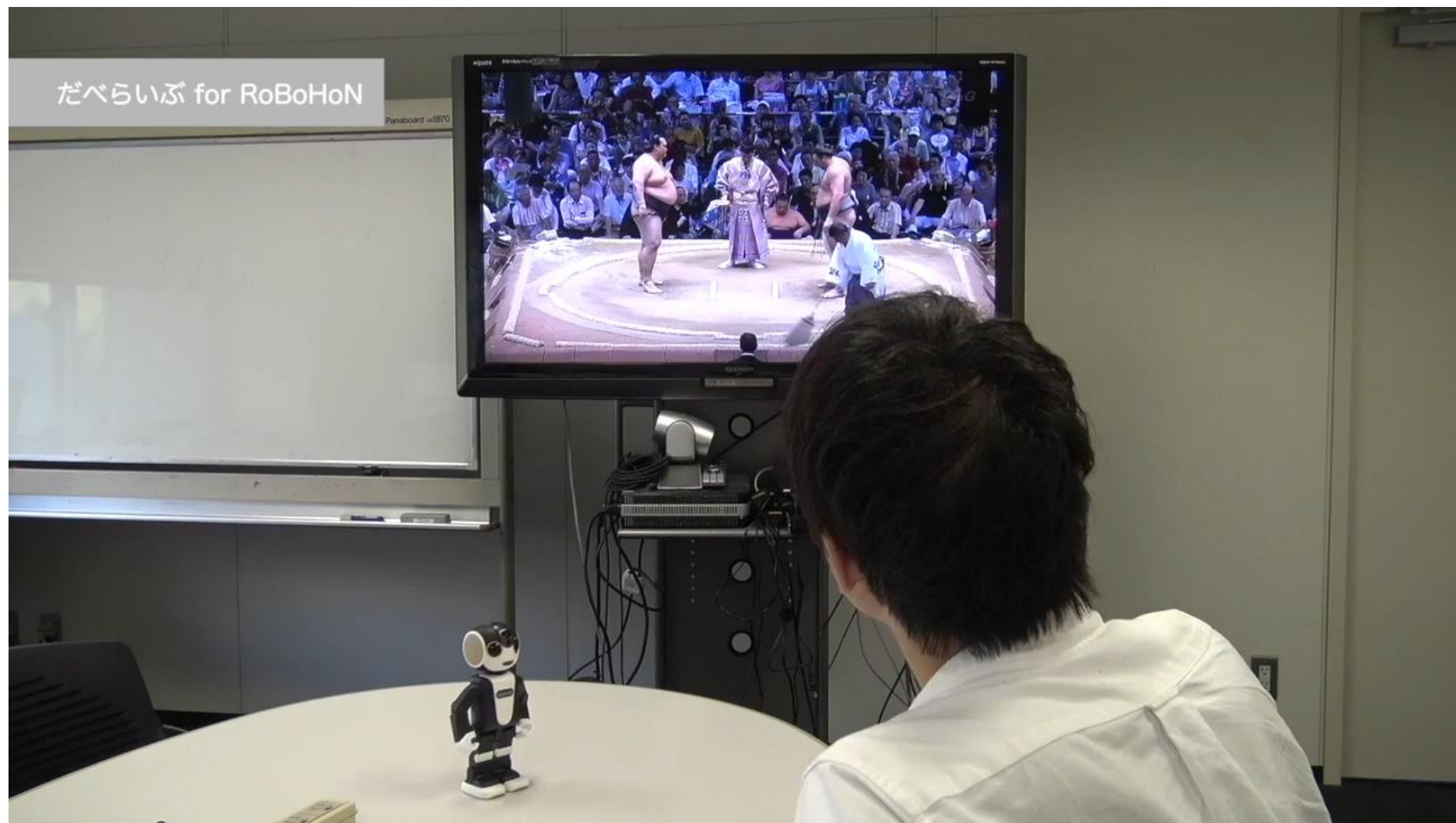


[Passenger enjoys "Hype Dlive"]

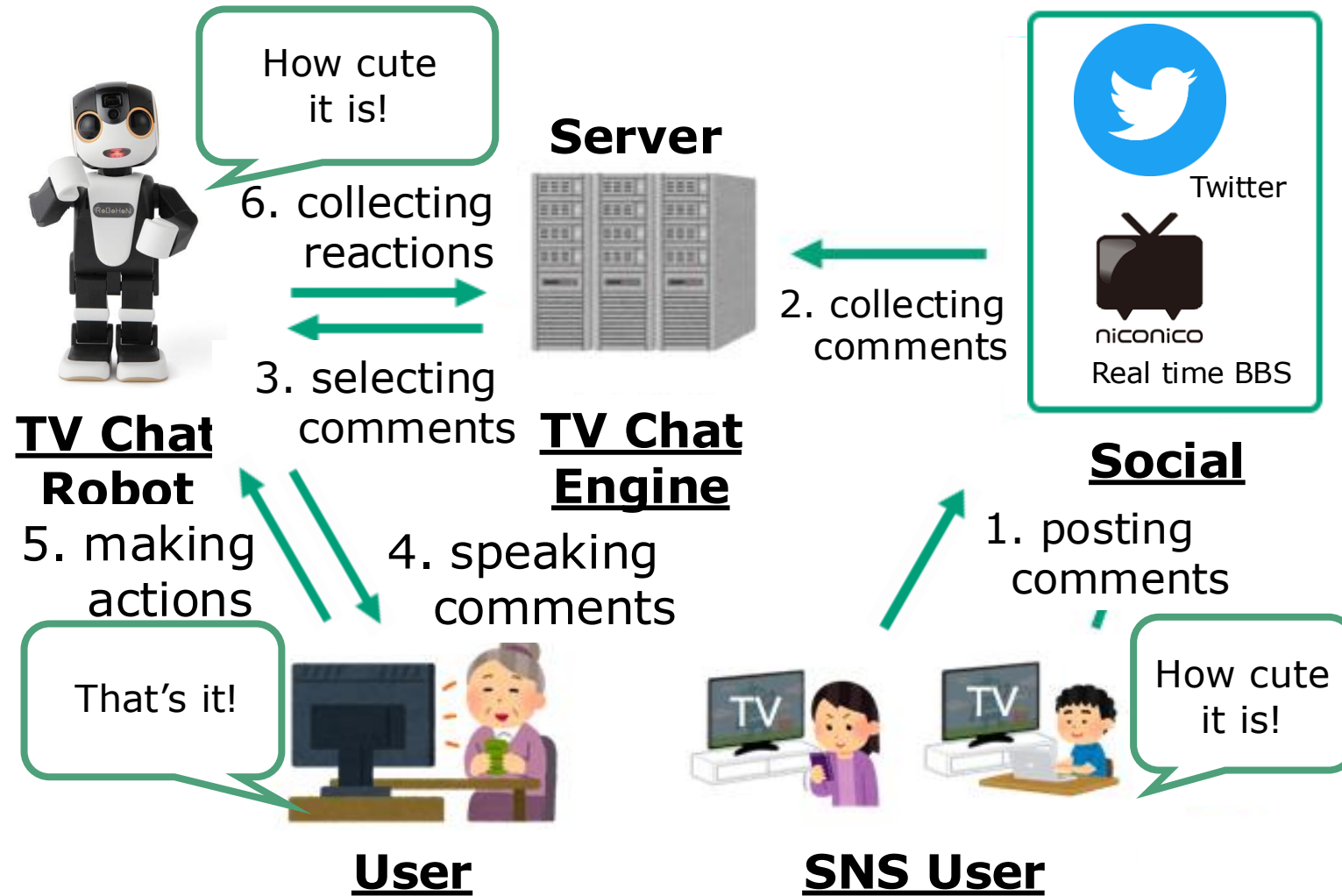


[Autonomous vehicle]

TV chat robot



Framework of TV chat robot



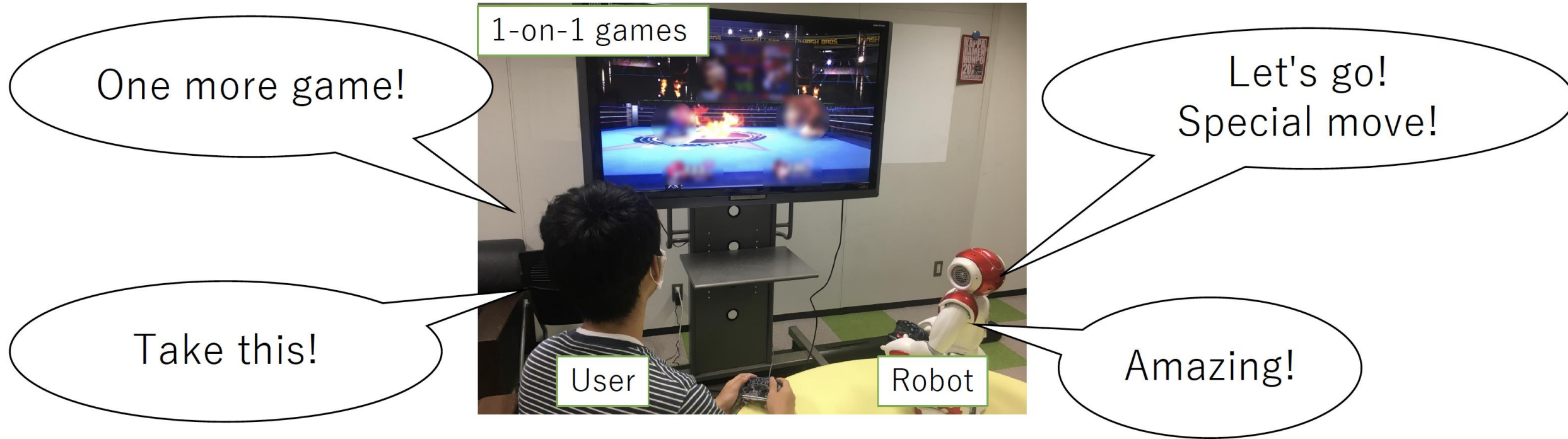
Atmosphere sharing : exciting scene

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Proposed Method :

Communication robot that plays video games together

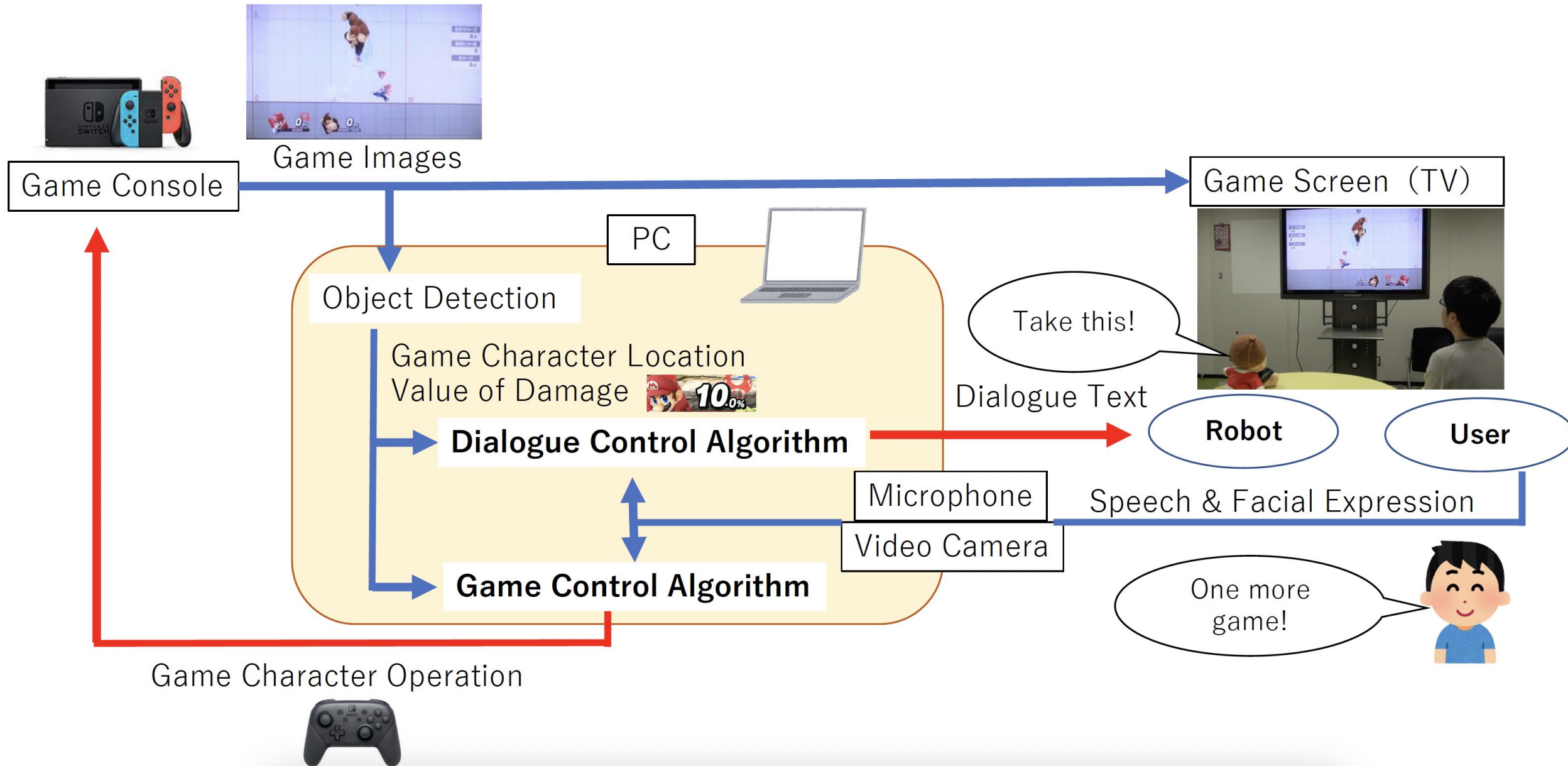


Fun playing video game with the communication robot



Increase motivation for daily-use of communication robot

System Overview of Proposed Robot



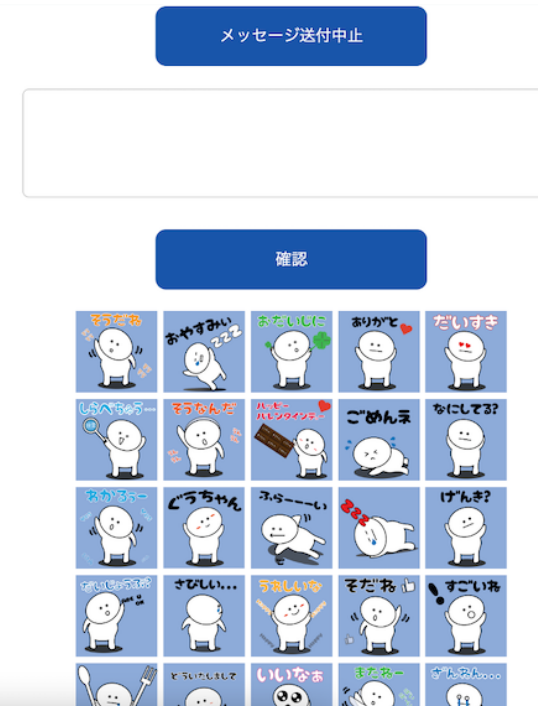
Digital divide solution for using SNS (LINE)

Communication agent for online smartphone support:

管理者画面1: ユーザリスト表示

管理者画面2: ユーザ詳細表示

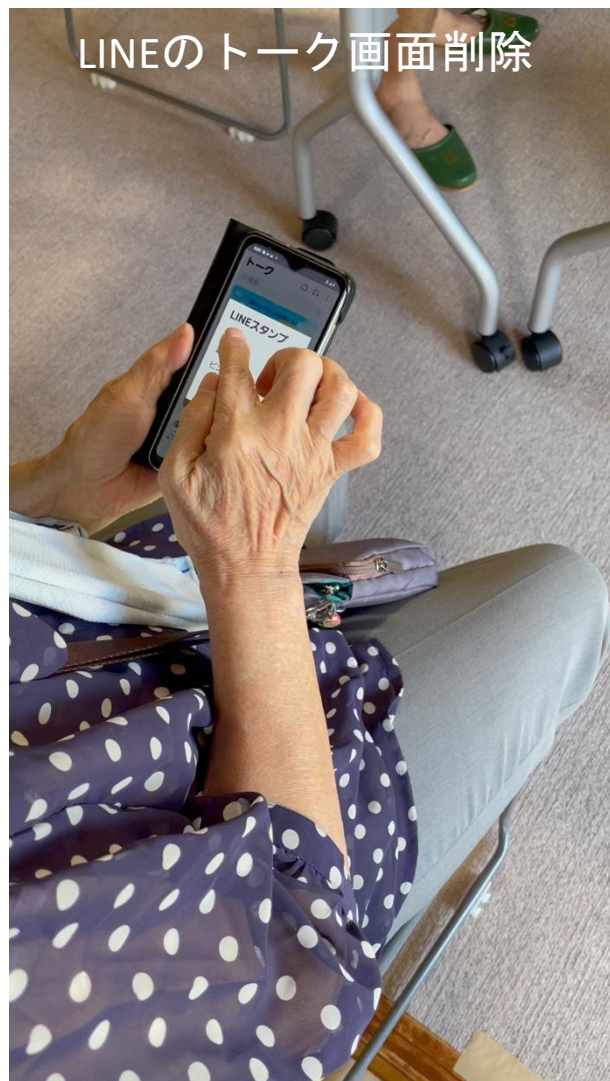
ユーザ画面: LINEチャット



[14] LINE, "LINE API", LINE Developers, <https://developers.line.biz/en/services/messaging-api/> (2023/02/03 閲覧).

[15] Kelley, J. F., "CAL – A Natural Language program developed with the OZ Paradigm: Implications for Supercomputing Systems", First International Conference on Supercomputing Systems, pp. 238-248, 1985.

Image of smart phone support:



Overview of this study

- **What:**

- To investigate the impact of avatars with **distinct visual characteristics** (muscular vs. skinny, male vs. female appearance) on the **robot's tactile perception interaction** warrants consideration

- **Comparison with related studies:**

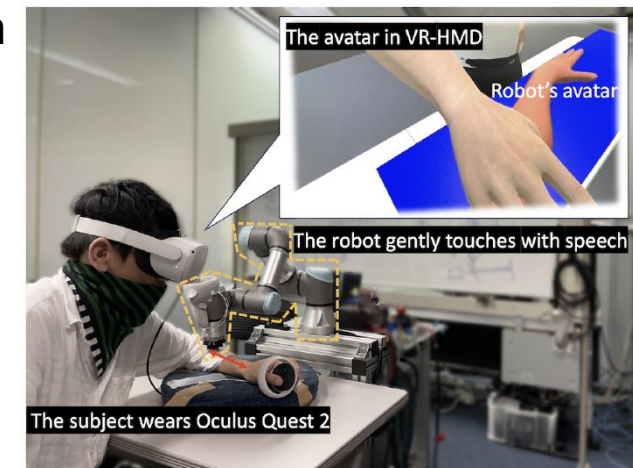
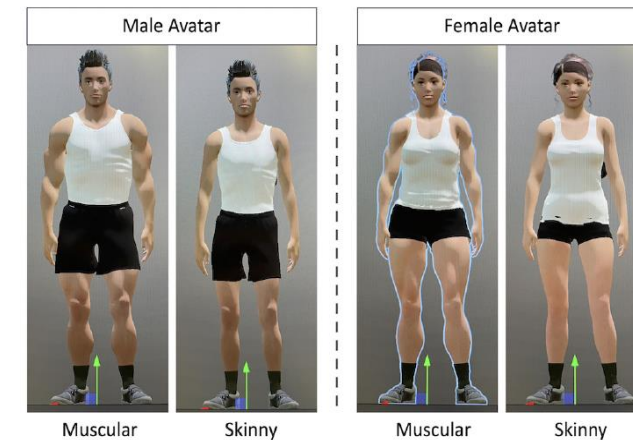
- Focus on the impact of avatars with distinct visual characteristics (**muscular vs. skinny, male vs. female appearance**)

- **Technical point:**

- By using **AR** technology to **change the appearance** of the robot arm itself with **different characteristics**

- **How to evaluate:**

- Godspeed Like Ability Scales & Scales along adjective pairs



Egg-laying bird robot

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Outline :

Tsundere Interaction

-Operant Conditioning: the learning to adapt to a **reward** or **aversive** stimulus (**punishment**), so as to voluntarily perform a certain behavior. It is a basic theory of behaviorist psychology.



You should
work harder!



+



Cheer! You know I like
hard-working person.

Induce users to be a comfortable
with behavior modification





谢谢 Thank you

✉ : t.sawabe@is.naist.jp

